

Maschke's Theorem and Its Proof

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Question 1 (Maschke's Theorem). *Let G be a finite group, and let*

$$\rho : G \longrightarrow \mathrm{GL}(V)$$

be a finite-dimensional representation of G over a field $\mathbb{F}$. If

$$\mathrm{char}(\mathbb{F}) \nmid |G|,$$

then ρ is completely reducible. More precisely, for every G -invariant subspace $W \subseteq V$, there exists a G -invariant subspace $U \subseteq V$ such that

$$V = W \oplus U.$$

Equivalently, every G -invariant subspace has a G -invariant complement.

Proof. The central idea of the proof is *averaging*. Since the group is finite, we can average an arbitrary linear projection over the group action and thereby turn it into a G -equivariant projection.

Step 1: Construct an arbitrary linear projection

Let $W \subseteq V$ be a G -invariant subspace. By linear algebra, we may choose a basis

$$\{w_1, \dots, w_m\}$$

of W , and extend it to a basis

$$\{w_1, \dots, w_m, u_1, \dots, u_k\}$$

of V . Define a linear map

$$\pi : V \longrightarrow W$$

by

$$\pi(w_i) = w_i, \quad \pi(u_j) = 0, \quad i = 1, \dots, m, \quad j = 1, \dots, k.$$

Then

$$\pi|_W = \text{id}_W, \quad \text{im}(\pi) = W, \quad \pi^2 = \pi.$$

Thus π is a projection onto W . However, π is not necessarily G -equivariant; it is merely a linear projection.

Step 2: Average the projection

Define a new linear map $\tilde{\pi} : V \rightarrow V$ by

$$\tilde{\pi}(v) = \frac{1}{|G|} \sum_{g \in G} \rho(g) \pi(\rho(g)^{-1}v), \quad v \in V.$$

The scalar $|G|^{-1}$ exists in $\mathbb{F}$, because $\text{char}(\mathbb{F}) \nmid |G|$.

Intuitively, for each $g \in G$, we first pull v back by $\rho(g)^{-1}$, project it onto W using π , push the result forward by $\rho(g)$, and then average over all group elements.

Step 3: Verify the key properties of $\tilde{\pi}$

(a) The image of $\tilde{\pi}$ is contained in W . For any $v \in V$ and $g \in G$, we have

$$\pi(\rho(g)^{-1}v) \in W.$$

Since W is G -invariant, applying $\rho(g)$ keeps the vector inside W . Hence

$$\rho(g) \pi(\rho(g)^{-1}v) \in W.$$

Therefore the finite sum also lies in W , and so

$$\text{im}(\tilde{\pi}) \subseteq W.$$

(b) The map $\tilde{\pi}$ restricts to the identity on W . Let $w \in W$. Since W is G -invariant, we have

$$\rho(g)^{-1}w \in W.$$

Because $\pi|_W = \text{id}_W$,

$$\pi(\rho(g)^{-1}w) = \rho(g)^{-1}w.$$

Therefore

$$\rho(g) \pi(\rho(g)^{-1}w) = \rho(g) \rho(g)^{-1}w = w.$$

Thus every term in the average equals w , so

$$\tilde{\pi}(w) = \frac{1}{|G|} \sum_{g \in G} w = w.$$

Consequently,

$$\tilde{\pi}|_W = \text{id}_W.$$

Together with $\text{im}(\tilde{\pi}) \subseteq W$, this implies

$$\text{im}(\tilde{\pi}) = W \quad \text{and} \quad \tilde{\pi}^2 = \tilde{\pi}.$$

(c) The map $\tilde{\pi}$ is G -equivariant. Let $h \in G$ and $v \in V$. Then

$$\begin{aligned} \tilde{\pi}(\rho(h)v) &= \frac{1}{|G|} \sum_{g \in G} \rho(g) \pi(\rho(g)^{-1} \rho(h)v) \\ &= \frac{1}{|G|} \sum_{g \in G} \rho(g) \pi(\rho(g^{-1}h)v). \end{aligned}$$

Now set $t = h^{-1}g$, so $g = ht$. As g ranges over G , so does t . Hence

$$\begin{aligned} \tilde{\pi}(\rho(h)v) &= \frac{1}{|G|} \sum_{t \in G} \rho(ht) \pi(\rho(t)^{-1}v) \\ &= \rho(h) \left(\frac{1}{|G|} \sum_{t \in G} \rho(t) \pi(\rho(t)^{-1}v) \right) \\ &= \rho(h) \tilde{\pi}(v). \end{aligned}$$

Therefore $\tilde{\pi}$ is a G -equivariant projection onto W .

Step 4: Take the kernel to obtain a G -invariant complement

Define

$$U = \ker(\tilde{\pi}) = \{u \in V \mid \tilde{\pi}(u) = 0\}.$$

Since $\tilde{\pi}$ is a projection with image W , linear algebra gives

$$V = \text{im}(\tilde{\pi}) \oplus \ker(\tilde{\pi}) = W \oplus U.$$

It remains only to check that U is G -invariant. If $u \in U$, then $\tilde{\pi}(u) = 0$. For any $g \in G$, using G -equivariance gives

$$\tilde{\pi}(\rho(g)u) = \rho(g)\tilde{\pi}(u) = \rho(g)0 = 0.$$

Thus $\rho(g)u \in U$, so U is G -invariant.

We have therefore found a G -invariant complement U of W , namely

$$V = W \oplus U.$$

□

Key Idea. The proof rests on four points:

1. Linear algebra provides an arbitrary projection $\pi : V \rightarrow W$.
2. Averaging over the finite group turns π into a G -equivariant projection $\tilde{\pi}$.
3. The change of variables in the group sum proves equivariance.
4. The kernel of $\tilde{\pi}$ is automatically a G -invariant complement of W .

Irreducible Representations of Finite Abelian Groups

Question 2. *Let G be a finite abelian group, and let $\mathbb{F}$ be an algebraically closed field such that*

$$\text{char}(\mathbb{F}) \nmid |G|.$$

Then every irreducible representation of G over $\mathbb{F}$ is one-dimensional.

Proof. Let (ρ, V) be an irreducible representation of G . Since G is abelian, for any $g, h \in G$ we have

$$\rho(g)\rho(h) = \rho(gh) = \rho(hg) = \rho(h)\rho(g).$$

Thus every operator $\rho(g)$ commutes with every operator $\rho(h)$. In particular, each $\rho(g)$ commutes with the whole image $\rho(G)$, so $\rho(g)$ is a G -module endomorphism of V .

Because V is irreducible and $\mathbb{F}$ is algebraically closed, Schur's lemma implies that every such endomorphism is scalar. Hence for each $g \in G$ there exists $\lambda_g \in \mathbb{F}$ such that

$$\rho(g) = \lambda_g \text{id}_V.$$

Therefore every one-dimensional subspace of V is G -invariant. If $\dim V > 1$, then V contains a nontrivial proper G -invariant subspace, contradicting irreducibility. So $\dim V = 1$. □

Characters

A one-dimensional representation is exactly a group homomorphism

$$\chi : G \longrightarrow \mathbb{F}^\times = \text{GL}(1, \mathbb{F}),$$

which is called a *character* of G . The set of all characters forms a group under pointwise multiplication:

$$\widehat{G} = \text{Hom}(G, \mathbb{F}^\times),$$

called the dual group of G .

The Cyclic Case

If $G = C_n = \langle x \mid x^n = 1 \rangle$, then every irreducible representation is determined uniquely by the image of the generator x . Let $\omega \in \mathbb{F}$ be a primitive n -th root of unity. For each $k = 0, 1, \dots, n-1$, define

$$\chi_k(x) = \omega^k.$$

Then

$$\chi_k(x^j) = \omega^{kj},$$

and these give all inequivalent irreducible representations of C_n . Hence

$$\widehat{C_n} = \{\chi_0, \chi_1, \dots, \chi_{n-1}\} \cong C_n.$$

The General Finite Abelian Case

By the structure theorem for finite abelian groups,

$$G \cong C_{n_1} \times C_{n_2} \times \cdots \times C_{n_r}.$$

Thus every irreducible representation of G is obtained by combining the characters of the cyclic factors. More precisely, if ω_i is a primitive n_i -th root of unity, then for $0 \leq k_i < n_i$ we have a character

$$\chi_{(k_1, \dots, k_r)}(x_1^{j_1}, \dots, x_r^{j_r}) = \prod_{i=1}^r \omega_i^{k_i j_i}.$$

Therefore:

1. The number of inequivalent irreducible representations is $n_1 n_2 \cdots n_r = |G|$.
2. The dual group satisfies

$$\widehat{G} \cong C_{n_1} \times \cdots \times C_{n_r} \cong G,$$

although this isomorphism is not canonical in general.

Orthogonality Relations

When $\mathbb{F} = \mathbb{C}$, the irreducible characters satisfy the usual orthogonality relation

$$\frac{1}{|G|} \sum_{g \in G} \chi_i(g) \overline{\chi_j(g)} = \delta_{ij}.$$

In particular, if $\chi \neq \mathbf{1}$ is a nontrivial character, then

$$\sum_{g \in G} \chi(g) = 0.$$

Relation with Maschke's Theorem

Maschke's theorem says that every finite-dimensional representation of G is completely reducible when $\text{char}(\mathbb{F}) \nmid |G|$. For a finite abelian group, the theorem above identifies the irreducible building blocks explicitly: they are precisely the characters. Hence every representation of a finite abelian group decomposes as a direct sum of characters:

$$V \cong \bigoplus_{\chi \in \hat{G}} m_{\chi} \chi,$$

where $m_{\chi} \in \mathbb{Z}_{\geq 0}$ is the multiplicity of χ . This is the representation-theoretic prototype of Fourier analysis on finite abelian groups.